Course Title

Management of AI Products

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Content: Experiential Learning Assignment / Capstone
Instructor: [Instructor Name]

Milestone 1 — Topic and Strategy Fix

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Working Title

PolicyGuard AI: An Intelligent Compliance AI Assistant for Insurance Underwriting Document Audit and Automated Policy Renewal.

Industry / Function

Insurance / Underwriting Operations / Compliance & Regulatory Audit / Policy Administration & Renewal.

Target User

- Underwriting Operations Analysts
- Compliance and Internal Audit Teams
- Policy Renewal and Servicing Teams
- Underwriting Managers
- Risk and Regulatory Compliance Officers

Current Workflow Problem

Insurance underwriting operations maintain thousands of policy-specific document folders, each containing a defined set of underwriting documents such as applications, risk assessments, inspection reports, endorsements, signed disclosures, and renewal correspondence. Regulatory and internal audit requirements mandate that these folders follow a prescribed folder structure, standardized file-naming conventions, and complete, accurate document content.

Today, underwriting operations human agents manually open and inspect each policy folder to check whether the folder structure is correct, whether file names follow naming standards, and whether the required documents are present, complete, and internally consistent. Because this review is done folder-by-folder and document-by-document, it takes days to months to discover compliance gaps, and issues are often found only when a third-party audit is already underway or when a policy is up for renewal.

There is no practical way to scale this review through additional manual effort. The volume of policy folders is growing continuously, naming and structural conventions vary across business lines and legacy systems, and content-level issues (missing signatures, mismatched policy numbers, outdated forms, inconsistent effective dates) require reading and reasoning over document content rather than simple file-system checks.

Why This Matters

This problem has significant business and regulatory impact because:

- Underwriting compliance issues are discovered late, often only during a third-party audit, increasing regulatory and reputational risk.
- Manual folder-by-folder review consumes significant analyst time that could otherwise be spent on underwriting judgment and risk analysis.
- Inconsistent folder structures and file names across business lines make it difficult to standardize audit readiness at scale.
- Policy renewals are delayed or processed with incomplete documentation because compliance gaps are not identified before the renewal decision.
- Audit findings and remediation effort are duplicated across similar folder or documentation issues that recur across policies.
- As experienced underwriting operations staff move roles or leave, institutional knowledge of “what good looks like” for a compliant folder is lost.

Improving the speed and consistency of underwriting document compliance checks can shorten audit preparation cycles, reduce regulatory exposure, accelerate policy renewal turnaround, and free underwriting staff to focus on higher-value risk decisions.

Proposed AI Role

AI-Powered Underwriting Compliance Copilot using Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Document AI/OCR for structured and unstructured document understanding.

The solution will:

- Scan policy-specific document folders and validate folder structure against the required compliance taxonomy.
- Validate file names against standardized naming conventions and flag mismatches or missing files.
- Read and reason over document content to detect missing signatures, expired or mismatched dates, incorrect policy numbers, and incomplete disclosures.
- Cross-reference document content across a folder to identify internal inconsistencies (e.g., risk assessment date preceding application date).
- Generate a folder-level audit-readiness report with specific, evidence-linked compliance findings for third-party audit preparation.
- Assess existing policyholders against renewal eligibility and documentation-completeness criteria, and generate a renewal recommendation with supporting evidence.

Why AI Instead of a Simpler Alternative

Traditional rule-based scripts or file-system validation tools can check for the presence of a file name pattern but cannot read or reason about document content, understand context across a folder, or recognize semantically equivalent documents that use different naming or formatting conventions across business lines and legacy systems.

A simple document repository, checklist, or dashboard can store and display documents but cannot understand whether the content within a document is complete, correct, or consistent with other documents in the same folder.

A RAG-based AI assistant can:

- Understand underwriting document types and their required content elements, not just file names.
- Reason across multiple documents in a folder to detect cross-document inconsistencies.
- Generalize compliance rules across business lines and legacy naming conventions rather than relying on rigid pattern matching.
- Summarize audit findings and renewal eligibility with references to the specific supporting documents.
- Scale consistently across thousands of policy folders, something manual human review cannot economically achieve.

Therefore, AI provides significantly greater value than rule-based scripts, static checklists, or manual folder review for this scale of underwriting compliance and renewal workflow.

Human-in-the-Loop Design

The system is designed as a decision-support and audit-preparation assistant rather than an autonomous compliance decision-maker or renewal authority.

Human reviewers will remain responsible for:

- Reviewing AI-flagged folder structure, file-naming, and content compliance findings before they are finalized.
- Validating AI-identified document inconsistencies and confirming whether they represent genuine compliance gaps.
- Approving the audit-readiness report submitted for third-party audit.
- Reviewing and approving or rejecting AI-generated policy renewal recommendations.
- Overriding AI findings when necessary and escalating ambiguous or high-risk cases to compliance or underwriting subject matter experts.

The AI system will provide supporting evidence, document references, and confidence indicators for every finding, while final compliance sign-off and renewal decisions will always remain with human underwriting and compliance staff.

Smallest Credible MVP

A web-based prototype that allows a user to point to a policy document folder (or upload a sample folder) and receive:

1. Folder structure compliance check against the required document taxonomy.
2. File naming convention validation with flagged mismatches.
3. Missing, duplicate, or mismatched document detection within the folder.
4. Content-level compliance flags (e.g., missing signatures, expired documents, inconsistent policy numbers or dates).
5. A folder-level audit-readiness score with evidence-linked findings.
6. An AI-generated policy renewal eligibility recommendation with supporting rationale.

The MVP focuses only on compliance detection, evidence generation, and renewal-eligibility insight for a single policy folder at a time, rather than full audit workflow automation or bulk remediation.

Build vs Buy vs API View

Recommendation: API-First Approach with RAG Architecture and Document AI/OCR APIs

Why:

- Existing LLM and Document AI/OCR APIs provide sufficient document understanding, reasoning, and summarization capability for an MVP.
- The primary value lies in encoding the organization's specific folder structure, naming conventions, and compliance rules, rather than building a custom foundation model.
- An API-first approach reduces development effort, implementation cost, and time-to-value.
- The team can focus on document ingestion, compliance-rule configuration, retrieval quality, and underwriting workflow integration.
- The architecture can later evolve toward a more customized or on-premises solution if business value is proven through pilot deployment, particularly given the sensitivity of policyholder data.

Main Risks

- Exposure of sensitive policyholder personal and financial data through AI processing and storage.
- Inaccurate or hallucinated compliance findings leading to incorrect audit or renewal conclusions.
- Over-reliance on AI-generated compliance findings without adequate human validation before audit submission or renewal approval.
- Inconsistent or incomplete historical folder structures across legacy business lines reducing detection accuracy during initial rollout.

Success Evidence

The solution will be considered successful if it demonstrates:

- Reduction in time required to complete a folder-level compliance check, from days/months to hours.
- Earlier detection of compliance issues, well ahead of third-party audit cycles.
- Faster and more consistent policy renewal turnaround for existing policyholders.
- Higher consistency in compliance findings across underwriting operations analysts.
- Positive feedback from underwriting, compliance, and audit teams on usability and trust in AI findings.

- Reduction in audit findings that were previously discovered only during third-party audits.

Team Recommendation

Proceed

The team recommends proceeding with an MVP focused on AI-assisted compliance checking and renewal-eligibility insight for underwriting document folders. The problem is business-relevant and regulator-facing, is supported by a large volume of existing policy documentation, has measurable time and risk-reduction value, and provides a realistic opportunity to demonstrate how AI can scale underwriting compliance and renewal decision-making while maintaining human oversight.